Sexual-Violence Victimization, Household Adversity, and Adolescent Substance Use and Social Withdrawal: A Design-Aware Machine-Learning Analysis of the 2023 National Youth Risk Behavior Survey

Eshal Minhaj[1]

[1] Independent Researcher, Bhopal, Madhya Pradesh, India

Correspondence: Eshal Minhaj, Independent Researcher, Bhopal, M.P., India. Email: [corresponding email]. ORCID: [ORCID iD].

Data availability: All analyses use the public-use 2023 National YRBS microdata (CDC). Complete code, the reproducible pipeline, and an interactive results dashboard are available at `github.com/eshalmin-basit/CSA-data-science-project` and `eshalmin-basit.github.io/CSA-data-science-project`.

Abstract

Background. Childhood sexual abuse (CSA) is associated with maladaptive outcomes extending well beyond the stereotypes attached to survivors. We test a coping-theoretic account — that substance use and social withdrawal after sexual trauma function as regulation strategies — against four pre-stated hypotheses in the newest nationally representative US adolescent sample.

Methods. Cross-sectional analysis of the 2023 National Youth Risk Behavior Survey (N = 20,103 students, grades 9–12; three-stage cluster design). Exposure was a composite of three sexual-violence items; household adversity was a 0–3 count of new-to-2023 ACE items. We combined survey-weighted prevalence estimation and cluster-robust logistic regression with (i) stratified 5-fold cross-validated prediction using exposure/adversity/context features only, comparing class weighting against SMOTE; (ii) TreeSHAP interpretation; (iii) potential-outcomes mediation analysis through persistent sadness/hopelessness; and (iv) a calibrated, fairness-audited screening prototype evaluated on a held-out test set with bootstrap confidence intervals, operating-point analysis, and decision-curve analysis. E-values quantified robustness to unmeasured confounding.

Results. Weighted exposure prevalence was 14.6% (21.6% female, 8.1% male). Exposed students showed higher weighted prevalence of all twelve substance outcomes, with prevalence ratios rising monotonically with substance severity (alcohol 2.17 → injection drug use 7.88), and lower school connectedness (0.76) and parental monitoring (0.85). Adjusted odds ratios were 2.19–3.19 across substance outcomes (all 95% CIs excluding 1; E-values 2.3–3.0). Household adversity showed strict dose–response gradients. Persistent sadness/hopelessness mediated 15–23% of substance associations and 46% of the connectedness association (all ACME CIs excluding 0). Cross-validated ROC-AUC reached 0.70–0.79 from trauma and demographic features alone; SMOTE collapsed sensitivity relative to class weighting (e.g., 0.13 vs 0.65 for binge drinking). The calibrated screening model achieved held-out AUC 0.76 (95% CI 0.72–0.80) for lifetime illicit drug use and identified 81% of currently-using students at a decision-curve-supported threshold, with subgroup AUC gaps ≤ 0.045.

Conclusions. All four hypotheses were supported. The severity-graded, distress-mediated pattern is consistent with trauma-driven coping rather than intrinsic risk-seeking. Trauma-informed screening with data schools already collect appears feasible, subject to prospective validation and strict ethical guardrails. Findings are associations from cross-sectional self-report and are framed as lower bounds, not causal effects.

Keywords: childhood sexual abuse; adverse childhood experiences; adolescent substance use; school connectedness; machine learning; causal mediation; risk screening; calibration; YRBS

Plain-Language Summary

This study asks a simple question: when teenagers who have experienced sexual violence drink, vape, use drugs, or pull away from the people around them, is that because they are "risk-taking" people — or because they are trying to cope with something unbearable? We tested five predictions in a survey of 20,103 US high school students, each designed so that the two explanations would give different answers.

- **Breadth (H1a).** If trauma drives coping, affected students should use *more of everything*, not just one or two "signature" substances. They did: every one of twelve substance outcomes was elevated.
- **Severity gradient (H1b).** This is the prediction that best separates the two explanations. A "risky personality" would raise all risky behaviors roughly equally. Coping that *escalates* predicts the biggest relative gaps for the rarest, most dangerous substances — the ones almost nobody touches without serious inner pressure. That is what we found: exposed students were about 2× more likely to drink alcohol, but 6–8× more likely to have used heroin, methamphetamine, or injection drugs.
- **Pathway (H2).** If these behaviors are coping with distress, then measured sadness should statistically *carry* part of the link — trauma leads to sadness, and sadness leads to substance use or withdrawal. It does: roughly a fifth of each substance association, and nearly half of the social-withdrawal association, runs through persistent sadness. The difference makes sense: withdrawing *is* an expression of sadness, while substances also do jobs (numbing, control, sleep) that a sadness question cannot capture.
- **Dose–response (H3).** Each *additional* hardship at home — a parent with a substance problem, a parent with mental illness, a parent in jail — should push every outcome up step by step, with no dips. It did, on every outcome we examined.
- **Predictability (H4).** If trauma burden is the real driver, a model told only *what happened to a student* — never how they behave, never their race or identity — should still identify most students at risk. It identified four out of five, and the model's own explanations rank the trauma variables above every demographic one.

All five predictions held. The picture that emerges is not one of reckless teenagers, but of young people managing pain with the tools available to them. Because this is a one-time anonymous survey, we can show these patterns travel together — we cannot prove cause and effect. But the practical message stands either way: asking a handful of questions about what a student has been through — questions schools can already ask — finds most of the students who need support, *before* the behavior becomes the problem, and points to mental-health care rather than punishment as the response.

---

1. Introduction

Adolescents who have experienced sexual violence carry an elevated burden of psychological distress, and a large clinical literature links childhood sexual abuse to later substance use, depression, and social difficulties (Hailes et al., 2019; Norman et al., 2012). Public discussion of CSA sequelae, however, tends to fixate on a narrow set of stereotyped outcomes. This study starts from a different premise: that the behaviors most strongly associated with trauma exposure — substance use across the full severity spectrum, and withdrawal from school and family connection — are better understood as coping mechanisms aimed at regaining stability and control (Khantzian, 1997). Taking that premise seriously changes what should be measured (the full severity gradient, not a signature behavior), what should be modeled (distress as mediator, not covariate), and what should be built (sensitivity-first supportive screening, not punitive flagging).

1.1 Theoretical framework and hypotheses

The self-medication hypothesis (Khantzian, 1997) holds that substances are selected and used to regulate otherwise intolerable affective states. Trauma-exposed adolescents experience chronic hyperarousal, intrusive distress, and shame; substances offer rapid, reliable, self-administered relief, and social withdrawal reduces exposure to evaluation and reminders. Under this account — and in contrast to a dispositional "risk-seeking" account — we pre-state four falsifiable hypotheses:

- **H1a (Breadth).** Sexual-violence exposure is associated with elevated prevalence of *every* substance outcome, not one or two signature behaviors, and the associations survive adjustment for demographics and household adversity.
- **H1b (Severity gradient).** The *relative* association strengthens with substance severity/rarity. Escalating-coping predicts that the rarer, more dangerous substances — used almost exclusively under strong regulation pressure — show the largest prevalence ratios. A generalized risk-taking disposition predicts no such ordering, since it elevates all risk behaviors comparably.
- **H2 (Mediation).** A measurable share of each exposure–outcome association is statistically mediated by psychological distress (persistent sadness/hopelessness; poor current mental health), and the mediated share is *larger* for social withdrawal than for substance use, since withdrawal is a direct expression of distress whereas substance use also serves regulation functions not captured by mood items.
- **H3 (Cumulative adversity).** Household adversity exhibits a monotonic dose–response with every outcome, replicating the ACE gradient (Felitti et al., 1998; Dube et al., 2003) in a 2023 nationally representative adolescent sample.
- **H4 (Predictability).** Exposure, adversity, and context features alone — excluding all behavioral and mental-health inputs — support cross-validated discrimination adequate for sensitivity-first screening (ROC-AUC ≥ 0.70 for substance outcomes), with victimization variables outranking demographics in feature attributions.

The 2023 National YRBS is the first cycle to carry household-ACE, school-connectedness, and parental-monitoring items alongside the long-standing violence and substance batteries (Brener et al., 2024), making these hypotheses jointly testable in one representative sample for the first time.

Contributions. (i) A design-aware characterization of the sexual-violence → substance-use severity gradient in the newest national YRBS cycle; (ii) the first dose–response analysis of the 2023 household-ACE items against the full substance battery; (iii) a formal potential-outcomes mediation test of the distress pathway; (iv) a predictive protocol that excludes on-pathway mediators, quantifies the class-imbalance decision, and explains itself via SHAP; (v) a calibrated, fairness-audited screening prototype with bootstrap uncertainty, operating-point and decision-curve evidence, and a public interactive implementation; and (vi) identification of school-level questionnaire-version missingness in the 2023 public-use file, of independent methodological value to other YRBS analysts.

2. Related Work

2.1 CSA and substance use

Hailes et al. (2019), in an umbrella review of 19 meta-analyses (559 primary studies), report pooled odds ratios of roughly 1.8–3.3 linking CSA to psychiatric and substance-misuse outcomes; Norman et al. (2012) find comparable magnitudes for physical abuse and neglect with evidence satisfying several causal-plausibility criteria. Simpson and Miller (2002) document strongly elevated CSA prevalence in clinical substance-using populations. Our adjusted estimates (2.19–3.19; Section 5.4) fall squarely in the predicted range in a current population-representative adolescent sample — itself a non-trivial replication, since most prior estimates come from adult retrospection or clinical samples.

2.2 Cumulative adversity and dose–response

The CDC-Kaiser ACE study established the graded exposure–outcome relationship for adult health-risk behavior (Felitti et al., 1998), and Dube et al. (2003) extended it to illicit drug initiation with 2- to 4-fold risk increases per additional ACE. The household-dysfunction items entered the national YRBS only in 2023; we provide their first dose–response characterization against the adolescent substance battery.

2.3 Self-medication and coping theory

Khantzian’s (1997) framework generates the breadth, gradient, and mediation predictions of Section 1.1. Direct mediation tests in nationally representative adolescent data are rare; most evidence is clinical, adult, or bivariate. Section 5.7 supplies the missing test with formal effect decomposition (Imai et al., 2010).

2.4 School connectedness

Connectedness is among the most replicated protective factors in adolescent health (Resnick et al., 1997; Steiner et al., 2019). We quantify both its deficit among victimized students and — new to this literature — the share of that deficit running through measured distress (46%).

2.5 Machine learning on adolescent risk surveys

Tree ensembles on YRBS suicidality report AUCs in the mid-0.70s to low 0.80s using broad behavioral feature sets. Three recurrent weaknesses motivate our protocol: mixing on-pathway mediators (or downstream behaviors) into features, inflating apparent performance while destroying interpretive value; applying SMOTE (Chawla et al., 2002) without comparison to class weighting; and omitting calibration, decision-analytic utility (Vickers & Elkin, 2006), and subgroup auditing (Obermeyer et al., 2019) despite deployment claims. Sections 4.3–4.9 address each directly.

3. Data

3.1 Survey design and sample

The National YRBS is a biennial cross-sectional school-based survey using a stratified three-stage cluster design — counties (primary sampling units), then schools, then classes — representative of US public and private high school students in grades 9–12 (Brener et al., 2024). In 2023, 155 of 311 sampled schools participated (school response rate 49.8%); 20,386 of 28,308 sampled students submitted questionnaires (student response rate 71.0%; overall response rate 35.4%), yielding N = 20,103 usable records after CDC data editing. The 2023 cycle was administered for the first time on electronic tablets with programmed skip logic. Survey design variables (`weight`, `stratum`, `psu`) accompany every record; weights calibrate the sample to national enrollment by sex, race/ethnicity, and grade.

3.2 Data preparation

The public-use Access database contains raw responses (table `XXHq`, questions Q1–Q107) and CDC-computed dichotomous variables (`XXHqn`, QN coding 1 = yes, 2 = no). Our open pipeline (`src/export_mdb.py`, `src/build_dataset.py`) exports both tables via ODBC, merges them 1:1 on record ID, and recodes to an analysis file of 50 variables plus design columns. Recoding rules: QN dichotomies map {1, 2} → {1, 0} with missing preserved; ordinal items retain CDC category order; age maps category codes to years; race/ethnicity uses the CDC 8-category bridged variable. We use CDC's QN dichotomies rather than re-deriving cutoffs, inheriting their logical-consistency edits. Two composites are constructed:

- `csa_exposure` = 1 if any of Q19 (ever physically forced intercourse), Q20 (forced sexual things by anyone, past 12 months), Q21 (sexual dating violence, past 12 months) is yes; 0 if all answered items are no; missing if all three are missing.
- `household_adversity_score` = count of Q100 (parent/guardian problematic alcohol/drug use), Q101 (parent/guardian severe mental illness), Q102 (parent/guardian incarceration); missing if any component is missing (components co-missing at $r \approx 0.96$, so this restriction costs < 2% beyond the block itself).

Variable mapping was validated line-by-line against the 2023 Data User's Guide codebook, and the composite prevalence (15.5% unweighted) was verified against independent derivation from the raw items.

3.3 Measures

Table 1. Analysis constructs, source items, and weighted prevalence.

Construct	YRBS item(s)	Weighted %
Sexual-violence exposure (composite)	Q19 ∪ Q20 ∪ Q21	14.6
— Ever forced intercourse	Q19	8.6
— Sexual violence, any perpetrator (12mo)	Q20	11.4
— Sexual dating violence (12mo)	Q21	5.9
Parent/guardian substance problem	Q100	24.8
Parent/guardian mental illness	Q101	28.5
Parent/guardian incarcerated	Q102	14.4
Persistent sadness/hopelessness	Q26	39.7
Poor current mental health (30d)	Q84	28.5
Attempted suicide (12mo)	Q29	9.5
Current vaping (30d)	Q36	16.8
Current alcohol (30d)	Q42	22.1
Current binge drinking (30d)	Q43	8.8
Current marijuana (30d)	Q48	17.0
Current cigarettes (30d)	Q33	3.5
Rx pain-medicine misuse (ever)	Q49	11.6
Any illicit drug (ever; CDC composite)	qnillict	9.9
Feels close to people at school	Q103	55.3
Parental monitoring	Q104	84.0

Terminology. The YRBS records neither age at first victimization nor perpetrator identity; `csa_exposure` therefore measures sexual-violence victimization reported in adolescence — a proxy for, not a clinical determination of, childhood sexual abuse. Under systematic under-disclosure this measure is misclassified toward "unexposed," and nondifferential misclassification of a binary exposure biases associations toward the null; we accordingly interpret prevalence and associations as lower bounds.

3.4 Analytic samples

Analyses use complete cases per estimand; no exposure or outcome is ever imputed (rationale in 3.5).

Table 2. Analytic-sample flow.

Analysis	Requirement	n
Full sample	—	20,103
Exposure prevalence	csa_exposure observed	17,627
Unadjusted associations	+ outcome observed	10,925–19,600 (per outcome)
Adjusted odds ratios	+ adversity, demographics	10,006–10,533
Prediction (CV)	outcome + exposure observed	10,838–17,381
Mediation	+ mediator, adjustment set	9,929–10,440
Screening prototype	all 13 intake items observed	6,039–6,409

3.5 Missing data: mechanism before method

We diagnosed mechanisms before selecting handling strategies (full evidence in repository, `docs/missing_data.md`). Four mechanisms:

1. **Questionnaire-version missingness (structural, school-level).** Students in schools sampled for both national and state surveys completed their state's questionnaire, which omits the 20 new-to-2023 national items (ACE block, connectedness, monitoring). Signature: within-block missingness correlations r = 0.88–0.96; 83–87% of PSUs (n ≥ 30) all-or-none missing; item response *rises* from 54% (Q98) to 78% (Q99), impossible under attrition. The receiving subsample matches the full sample within 0.7 weighted percentage points on every audited construct (e.g., exposure 14.8% vs 14.1%; vaping 16.9% vs 16.5%); affected variables are analyzed in the subsample that received them.
2. **Within-questionnaire attrition** (97% response at Q48 → 65% at Q78), treated as missing-at-random given demographics.
3. **Skip-pattern missingness** from tablet branching, resolved by CDC logical edits in the QN variables.
4. **Sensitive-item nonresponse** on the sexual-violence items (14–22%), potentially value-related; exposure is never imputed and associations are framed as lower bounds.

Multiple imputation is deliberately not applied to block-missing variables: imputing 44% of a variable whose absence is a school-level design artifact would manufacture within-school variance without information gain.

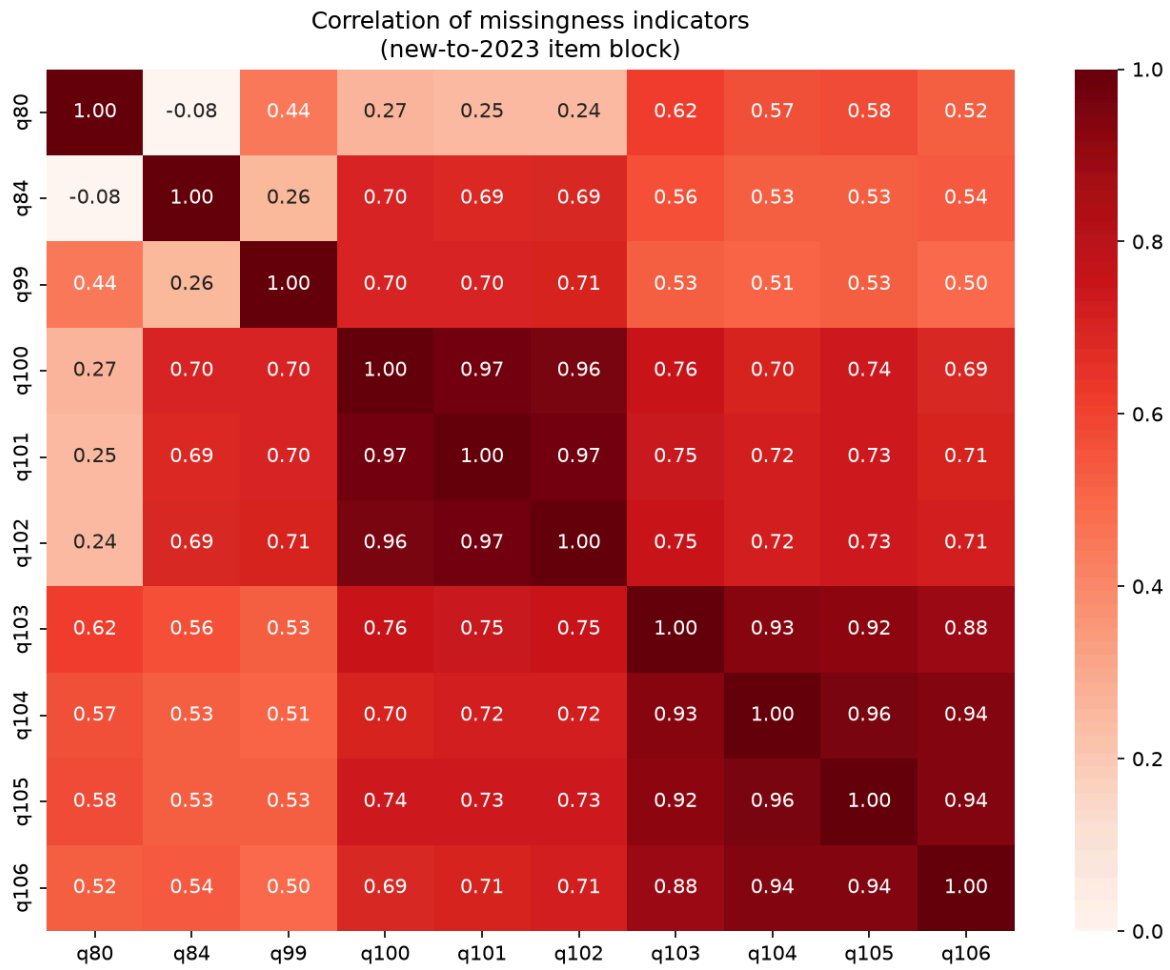

Figure 5. *Correlation of item-missingness indicators within the new-to-2023 item block. Near-unity within-block correlations identify school-level questionnaire-version missingness rather than individual nonresponse.*

3.6 Ethics

All data are de-identified public-use records; secondary analysis requires no IRB review. Reporting commitments: risk-factor language throughout, no causal claims; results framed to motivate protection and support, never stigma; the screening prototype framed strictly as a prioritization aid for supportive, non-punitive outreach, with prospective validation, IRB oversight, and community consent named as deployment preconditions.

4. Methods

4.1 Design-aware estimation

Weighted prevalence: $\hat{p} = \Sigma_i w_i y_i / \Sigma_i w_i$ over non-missing respondents. Adjusted associations use weighted logistic regression with weights normalized to the analytic n ($\tilde{w}_i = w_i \cdot n/\Sigma w_i$):

> $\text{logit}\, P(Y_i = 1) = \beta_0 + \beta_1 \cdot CSA_i + \beta_2 \cdot ADV_i + \gamma' X_i$

with ADV the 0–3 adversity count and X = {age, grade, sex, race/ethnicity (ref. White)}. Standard errors are cluster-robust (sandwich) on PSU. This approximates Taylor-linearized design-based variance while ignoring stratification gains (generally conservative); we report it as an approximation to full survey-package estimation.

4.2 Performance metric definitions

With TP/FP/TN/FN the confusion-matrix cells at threshold t:

- Sensitivity (recall) = TP/(TP+FN); specificity = TN/(TN+FP); precision (PPV) = TP/(TP+FP); F1 = 2·(precision·recall)/(precision+recall).
- ROC-AUC: probability a random positive is scored above a random negative; the area under {(FPR(t), TPR(t))}.
- PR-AUC: average precision, $\Sigma_n (R_n - R_{n-1}) \cdot P_n$; its no-skill baseline is the outcome prevalence, making it the informative summary under imbalance.
- Brier score = $(1/n)\Sigma_i(\hat{p}_i - y_i)^2$, proper for calibrated probability quality; calibration curves plot observed frequency against predicted probability in quantile bins.

4.3 Model choice and rationale

Each statistical task in this study has a different job, and each model was chosen for that job rather than for novelty:

- **Survey-weighted logistic regression with cluster-robust errors** (inference, §4.1). The estimand is a population association — "how much more likely is the outcome among exposed students, nationally?" — which requires design weights and clustering-aware uncertainty, not predictive machinery. Odds ratios are the field-standard effect measure, directly comparable to the meta-analytic literature in Section 2.
- **L2-regularized logistic regression** (prediction baseline). The strongest simple hypothesis: risk is an additive combination of exposures. It is fast, stable, produces probabilities that are nearly calibrated out of the box, and its coefficients are auditable as odds ratios — properties a screening deployment values more than raw accuracy.
- **Gradient-boosted trees (XGBoost; Chen & Guestrin, 2016)** (prediction challenger). The consistent state of the art for tabular data; captures non-linearities and interactions (e.g., exposure × unstable housing) that a linear model cannot, and handles missing values natively, which matters given the structural missingness of Section 3.5. Deep learning was not considered: with ~6,000–17,000 rows of low-dimensional tabular features it offers no expected accuracy advantage while sacrificing the interpretability a clinical audience requires.
- **Why both, always.** The logistic-versus-XGBoost gap is itself a diagnostic. If the ensemble wins, meaningful interaction structure exists; if the linear model matches it — as it did here (Table 6) — the signal is predominantly additive, and the transparent model earns the deployment recommendation on equal performance. Reporting only the more complex model would have obscured that finding.
- **Platt-calibrated XGBoost** (screening, §4.7). A screening threshold is a policy decision expressed in probability units; that requires "20% risk" to mean 20% observed frequency. Raw boosted-tree scores are not calibrated, so probabilities are rescaled by Platt's method (Platt, 1999) inside the training folds and verified on the test set (Brier scores; Figure 4b).

- **TreeSHAP (Lundberg & Lee, 2017)** (interpretation, §4.5). Chosen over permutation importance because it yields exact, per-student, signed attributions under the tree structure — enabling both the global ranking in Section 5.6 and the fairness-relevant check that demographic features do not dominate.
- **Probit GLMs within the Imai–Keele–Tingley framework** (mediation, §4.6). Mediation with a binary mediator and binary outcome requires effect decomposition on the potential-outcomes scale, not the product-of-coefficients shortcut, which is biased in non-linear models; the quasi-Bayesian algorithm (Imai et al., 2010) provides valid uncertainty for the ACME/ADE quantities of interest.

4.3.1 Predictive modeling protocol

For each of eight outcomes we fit an L2-regularized logistic regression and gradient-boosted trees (XGBoost; Chen & Guestrin, 2016). Features comprise only exposure/adversity/context — sexual-violence exposure, household adversity, physical dating violence, school and electronic bullying, witnessed community violence, unstable housing, basic-needs support — plus demographics. **Psychological mediators are excluded by design:** they lie on the hypothesized causal pathway (Section 1.1), and conditioning on them would launder exposure signal into apparent "mental-health" predictors, inflate apparent performance, and invalidate H4's interpretation. Rows are complete-case on {outcome, exposure}; remaining covariate missingness is median-imputed with missingness indicators in the logistic pipeline and passed natively to XGBoost. Evaluation: stratified 5-fold cross-validation, all preprocessing fit within training folds; we report mean ± SD across folds.

4.4 Class imbalance

Prevalence spans 4.1% (cigarettes) to 46.2% (low connectedness). We compare inverse-prevalence class weighting (logistic `class_weight='balanced'`; XGBoost `scale_pos_weight` = N_-/N_+) against SMOTE (Chawla et al., 2002) applied strictly inside training folds, holding all else fixed. Sensitivity at the default threshold is the headline comparison, matching the screening use-case.

4.5 Interpretability

TreeSHAP (Lundberg & Lee, 2017) provides exact additive attributions φ_{ij} with $\Sigma_j\varphi_{ij} = f(x_i) - E[f(X)]$; we report mean |SHAP| rankings and beeswarm distributions from 3,000-student samples per outcome.

4.6 Causal mediation

Under potential outcomes (Imai et al., 2010), with exposure T, mediator M, outcome Y:

> $\text{ACME}(t) = E[Y(t, M(1)) - Y(t, M(0))]$ (indirect effect) $\text{ADE}(t) = E[Y(1, M(t)) - Y(0, M(t))]$ (direct effect) Total effect = ACME(t) + ADE(1−t); proportion mediated = ACME/Total.

Mediator and outcome models are probit GLMs adjusted for age, grade, sex, race/ethnicity, and household adversity; uncertainty via quasi-Bayesian Monte Carlo with 500 parameter draws. Identification requires sequential ignorability — no unmeasured exposure–outcome, exposure–mediator, or mediator–outcome confounding, with mediator temporally after exposure — which cross-sectional data cannot verify. Estimates are reported as consistency checks for H2, not causal proof.

4.7 Screening prototype

Thirteen intake-collectable inputs (the victimization/adversity/context set plus age, grade, sex). **Race/ethnicity and sexual identity are excluded as inputs and retained as audit dimensions.** Protocol: stratified 80/20 train/test split; randomized hyperparameter search for XGBoost (30 candidates, 5-fold CV, ROC-AUC objective) over n_estimators ∈ [150, 700], max_depth ∈ [2, 5], learning rate ∈ logU(0.01, 0.2), subsample and colsample ∈ {0.7, 0.8, 0.9, 1.0}, min_child_weight ∈ [1, 19], λ ∈ logU(0.1, 10); Platt calibration (Platt, 1999) via 5-fold CalibratedClassifierCV on training data; logistic baseline fit on raw features so coefficients export as odds ratios. All reported metrics come from the untouched test set, with nonparametric bootstrap 95% CIs (1,000 resamples of test predictions). Operating points are sensitivity-first: the largest threshold achieving ≥ 90% and ≥ 80% recall.

4.8 Decision-curve analysis

Net benefit at threshold probability p_t (Vickers & Elkin, 2006):

> NB(p_t) = TP/n − (FP/n) · p_t/(1 − p_t)

compared against screen-everyone (NB_all = π − (1−π)·p_t/(1−p_t), π = prevalence) and screen-no-one (NB = 0) over p_t ∈ [0.02, 0.60].

4.9 Fairness audit

At the deployed operating point we report subgroup n, prevalence, AUC, sensitivity, false-positive rate, and precision by sex and race/ethnicity (groups with ≥ 150 test students and ≥ 20 positives), reporting residual gaps openly (Obermeyer et al., 2019).

4.10 Sensitivity analyses

(i) Weighted vs unweighted prevalence deltas for all key constructs; (ii) received-block vs no-block subsample comparability (Section 3.5); (iii) E-values (VanderWeele & Ding, 2017) for adjusted estimates: for common outcomes we first convert OR to an approximate risk ratio RR = √*OR, then E = RR* + √(RR*(RR*−1)) — the minimum strength of association an unmeasured confounder would need with both exposure and outcome to fully explain the estimate.

4.11 Software and reproducibility

Python 3.12.2; pandas 2.3.3, numpy 2.4.6, scikit-learn 1.8.0, xgboost 3.3.0, shap 0.52.0, statsmodels 0.14.6, imbalanced-learn 0.14.2. Fixed seed (42) for all splits, searches, and simulations. The full pipeline from raw Access database to every table below is reproducible from the repository; the dashboard renders the same precomputed artifacts.

5. Results

5.1 Sample and prevalence

Weighted sexual-violence exposure prevalence was 14.6% — 21.6% among female and 8.1% among male students; highest among American Indian/Alaska Native (16.5%) and multiracial students (17.5–17.7%).

Population-level burden was substantial: 39.7% persistent sadness/hopelessness, 28.5% poor current mental health, 9.5% past-year suicide attempt. Weighted and unweighted estimates differed by ≤ 5 percentage points throughout (sensitivity analysis i).

5.2 Breadth and severity gradient (H1a, H1b)

Table 3. Weighted prevalence by exposure status, ordered by prevalence ratio (PR).

Outcome	Exposed %	Unexposed %	PR
Alcohol (30d)	41.8	19.3	2.17
Marijuana (30d)	35.6	14.2	2.51
Rx pain-med misuse (ever)	24.2	8.9	2.73
Vaping (30d)	39.1	13.4	2.93
Binge drinking (30d)	21.6	6.7	3.21
Any illicit drug (ever)	24.5	7.5	3.28
Cigarettes (30d)	9.8	2.5	3.92
Inhalants (ever)	15.6	3.9	3.97
Heroin (ever)	5.5	0.9	6.32
Cocaine (ever)	9.0	1.4	6.38
Ecstasy (ever)	10.4	1.5	7.14
Methamphetamine (ever)	7.2	1.0	7.47
Injection drug use (ever)	4.7	0.6	7.88
Feels close to people at school	43.4	57.4	0.76
Parental monitoring	72.9	85.9	0.85
Persistent sadness/hopelessness	72.1	33.3	2.16
Considered suicide (12mo)	48.0	14.7	3.27
Attempted suicide (12mo)	28.7	5.8	4.96

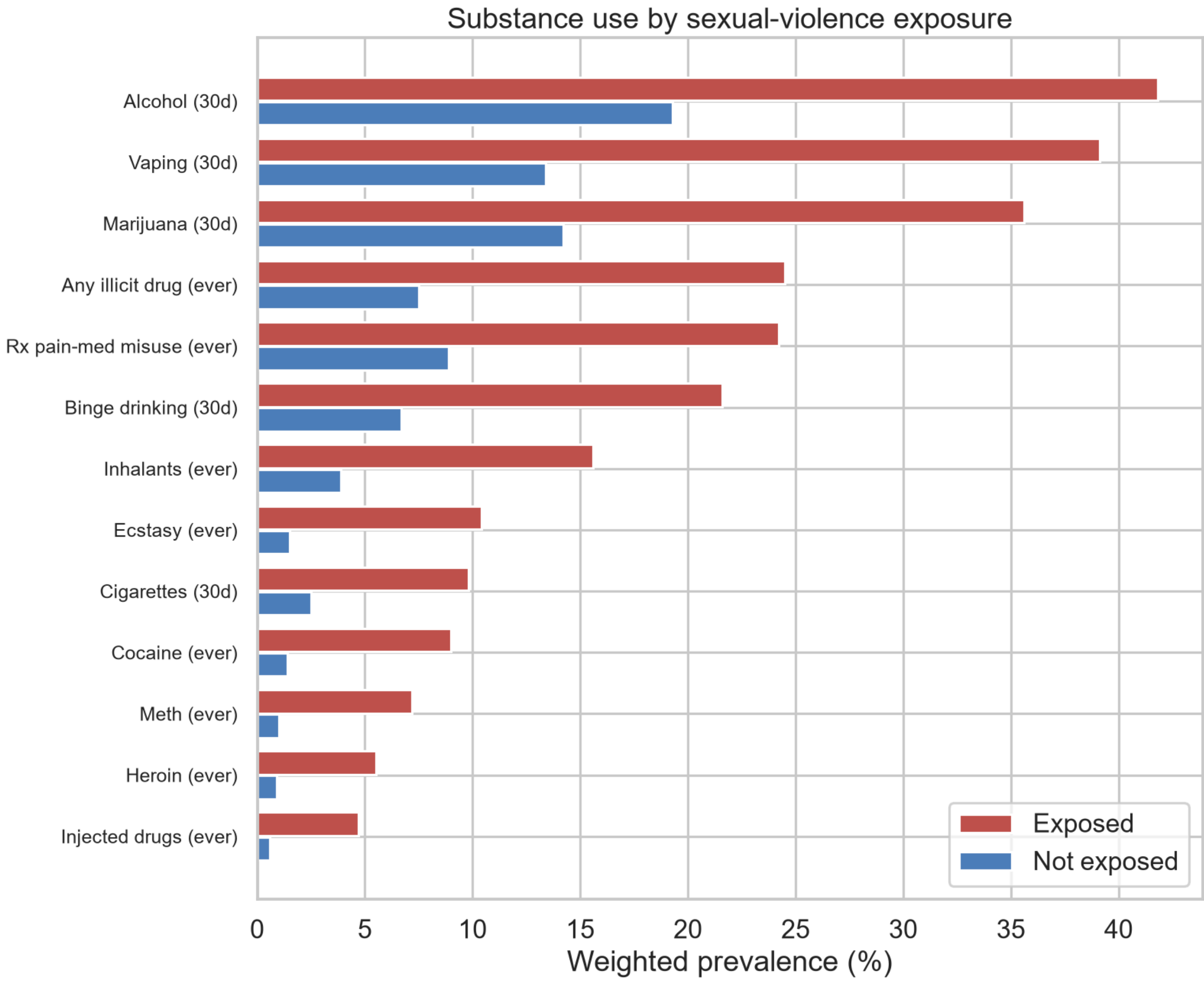

Figure 1. *Weighted prevalence of substance-use outcomes among students reporting sexual-violence victimization versus unexposed students, 2023 National YRBS.*

Every substance outcome is elevated (H1a; Figure 1), and the PR ordering tracks severity almost perfectly — common/social substances 2.2–2.9, cigarettes and inhalants ≈ 4, and the rarest, most dangerous substances 6.3–7.9 (H1b). Both connection measures fall below parity, the withdrawal signature. Among exposed students, 72.1% report persistent sadness and 28.7% a past-year suicide attempt.

5.3 Household adversity dose–response (H3)

Table 4. Weighted prevalence by adversity score.

Score	n	Any illicit %	Vaping %	Binge %	Sadness %	Suicide attempt %
0	7,024	5.8	10.5	6.4	27.1	4.3
1	2,868	10.7	18.7	8.2	53.5	11.8
2	1,936	18.2	31.1	15.1	65.2	19.7
3	1,198	24.8	37.2	15.6	71.4	26.3

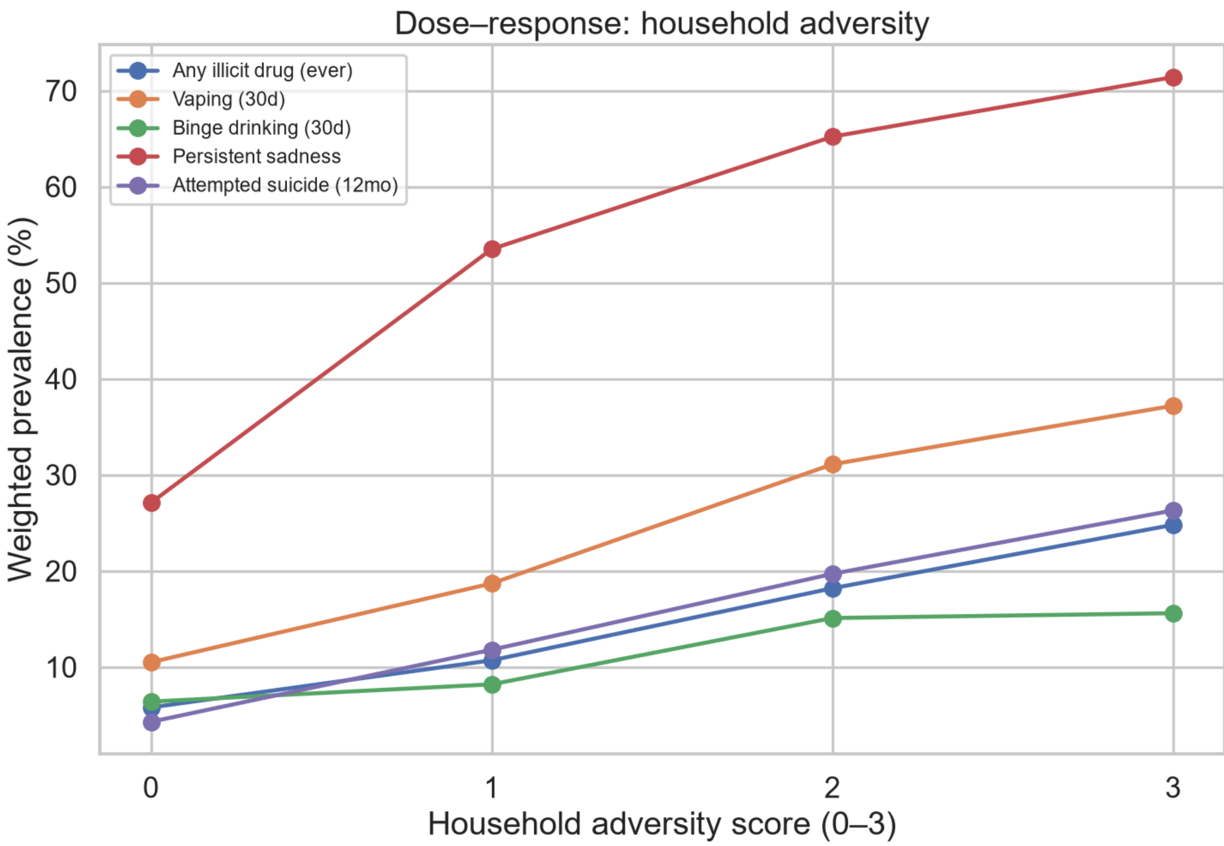

Figure 2. *Weighted prevalence of selected outcomes across the household adversity score (0–3). Every examined outcome rises monotonically.*

Strict monotonicity holds for every outcome examined (Figure 2) — the ACE gradient (Felitti et al., 1998; Dube et al., 2003) reproduced in the 2023 adolescent sample. H3 supported.

5.4 Adjusted associations and confounding robustness (H1a)

Table 5. Survey-weighted logistic regression (PSU cluster-robust 95% CIs); each model adjusts for age, grade, sex, race/ethnicity, and the other exposure. E-values per Section 4.10.

Outcome	CSA-proxy OR (95% CI)	E-value (CI bound)	Adversity OR/point (95% CI)	n
Any illicit drug (ever)	3.19 (2.50–4.06)	2.97 (2.54)	1.64 (1.48–1.83)	10,006
Vaping (30d)	2.89 (2.42–3.45)	2.79 (2.49)	1.62 (1.47–1.79)	10,152
Binge drinking (30d)	2.84 (2.34–3.45)	2.76 (2.43)	1.23 (1.09–1.39)	10,109
Cigarettes (30d)	2.67 (1.82–3.91)	2.65 (2.04)	1.74 (1.47–2.06)	10,533
Marijuana (30d)	2.31 (1.93–2.78)	2.41 (2.12)	1.62 (1.44–1.81)	10,489
Rx pain-med misuse (ever)	2.19 (1.75–2.73)	2.32 (1.98)	1.52 (1.37–1.69)	10,502
Low school connectedness	1.42 (1.24–1.62)	1.67 (1.47)	1.27 (1.17–1.38)	10,490

All intervals exclude 1; exposure and adversity contribute independently everywhere. The E-values indicate that explaining away the illicit-drug association would require an unmeasured confounder associated with both exposure and outcome at RR ≈ 3.0 (≥ 2.5 to move the CI to null) — stronger than most plausible candidates net of the adversity, demographic, and context variables already in the model.

5.5 Prediction (H4)

Table 6. Stratified 5-fold CV (mean ± SD); trauma/context + demographic features only; class-weighted models.

Outcome (prev.)	Model	ROC-AUC	Sensitivity	PR-AUC
Cigarettes (4.1%)	Logistic	0.789 ± 0.019	0.703 ± 0.034	0.175 ± 0.030
	XGBoost	0.754 ± 0.010	0.566 ± 0.040	0.152 ± 0.021
Binge drinking (9.2%)	Logistic	0.773 ± 0.014	0.690 ± 0.035	0.266 ± 0.026
	XGBoost	0.747 ± 0.020	0.645 ± 0.039	0.246 ± 0.021
Vaping (18.1%)	Logistic	0.766 ± 0.010	0.666 ± 0.020	0.429 ± 0.011
	XGBoost	0.758 ± 0.006	0.662 ± 0.013	0.416 ± 0.012
Any illicit ever (11.3%)	Logistic	0.760 ± 0.010	0.660 ± 0.028	0.333 ± 0.009
	XGBoost	0.739 ± 0.009	0.617 ± 0.018	0.307 ± 0.017
Marijuana (17.9%)	Logistic	0.758 ± 0.006	0.662 ± 0.015	0.408 ± 0.011
	XGBoost	0.750 ± 0.010	0.661 ± 0.023	0.398 ± 0.011
Alcohol (21.9%)	Logistic	0.727 ± 0.005	0.651 ± 0.018	0.419 ± 0.006
	XGBoost	0.726 ± 0.004	0.660 ± 0.012	0.425 ± 0.007
Rx misuse (11.9%)	Logistic	0.715 ± 0.015	0.586 ± 0.029	0.285 ± 0.017
	XGBoost	0.701 ± 0.012	0.558 ± 0.020	0.283 ± 0.019
Low connectedness (46.2%)	Logistic	0.620 ± 0.018	0.542 ± 0.014	0.565 ± 0.010
	XGBoost	0.606 ± 0.015	0.541 ± 0.009	0.553 ± 0.015

Three findings. **First**, all seven substance outcomes clear the H4 threshold (AUC 0.70–0.79) from exposure information alone. **Second**, logistic regression matches or exceeds tuned XGBoost on every substance outcome — the risk signal is predominantly additive, which favors the interpretable model for deployment. **Third**, the imbalance comparison is decisive: replacing class weighting with SMOTE collapses XGBoost sensitivity at the default threshold — 0.133 vs 0.645 (binge drinking), 0.037 vs 0.566 (cigarettes), 0.059 vs 0.558 (Rx misuse), 0.108 vs 0.617 (any illicit) — with no compensating AUC gain. On sparse binary survey features, SMOTE's synthetic interpolants concentrate mass near the minority-class centroid and shift the decision surface away from sensitivity; simple reweighting preserves it. Low school connectedness is markedly harder (AUC 0.62) than any substance outcome: withdrawal depends on school-contextual factors beyond measured trauma variables.

5.6 Feature attributions

Across all eight outcomes, mean |SHAP| rankings place the victimization/adversity block above every demographic feature: physical dating violence, household adversity, sexual-violence exposure, and witnessed community violence occupy the top positions, with basic-needs support acting protectively (Figure 3). The models predict from *what happened to the student*, not from who the student demographically is — completing H4.

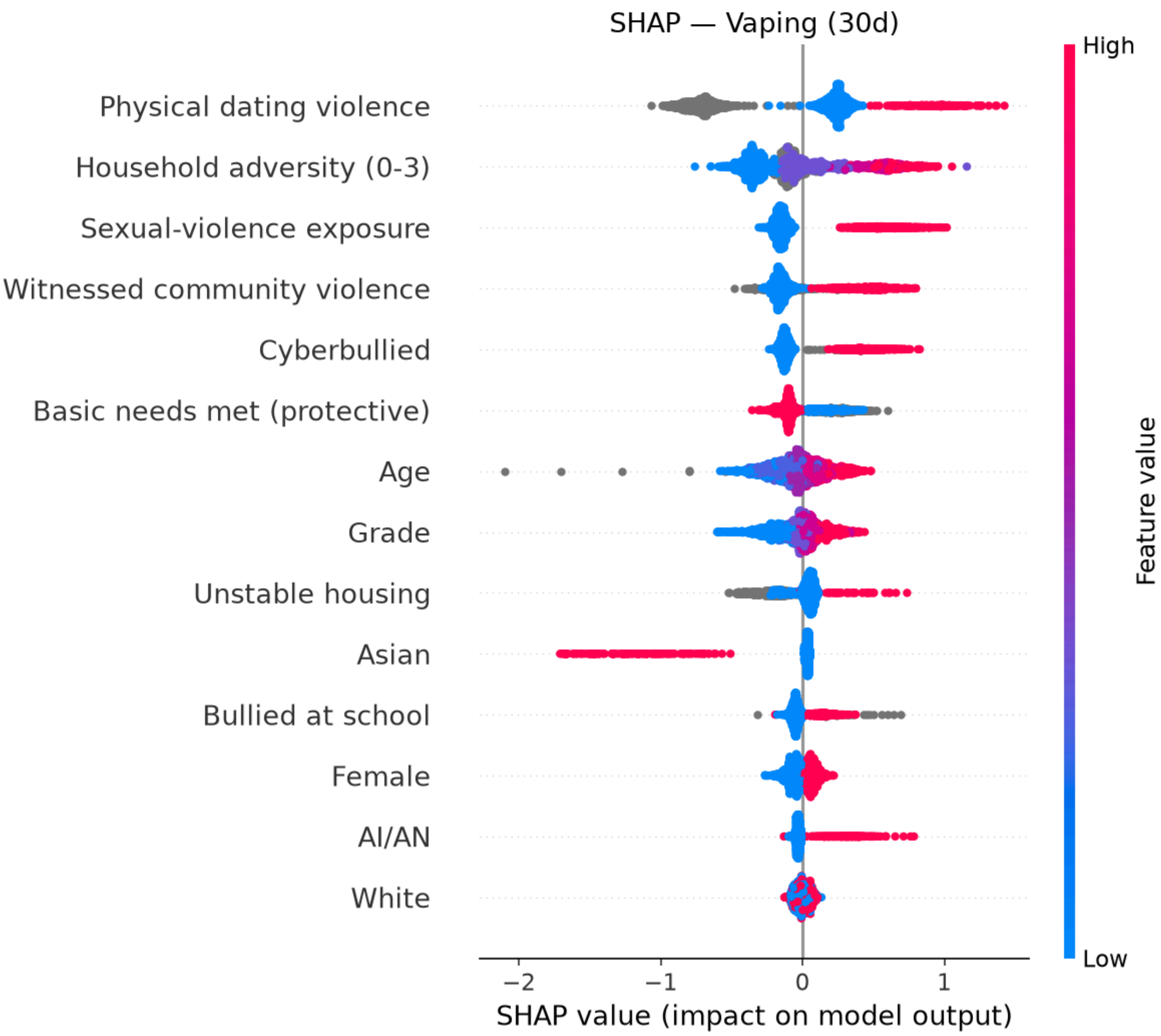

Figure 3. *SHAP beeswarm for the current-vaping model (class-weighted XGBoost, 3,000-student sample). Each point is one student; horizontal position is the feature's contribution to that student's predicted risk. Victimization and adversity features dominate; corresponding plots for the other seven outcomes are in the repository.*

5.7 Mediation (H2)

Table 7. Effect decomposition, mediator = persistent sadness/hopelessness (probit, 500 quasi-Bayesian draws; risk-difference scale).

Outcome	Total effect	ACME (95% CI)	ADE (95% CI)	Prop. mediated (95% CI)	n
Low school connectedness	0.085	0.039 (0.033, 0.046)	0.045 (0.020, 0.071)	0.46 (0.35, 0.67)	10,430
Marijuana (30d)	0.134	0.031 (0.025, 0.037)	0.102 (0.081, 0.124)	0.23 (0.19, 0.29)	10,428
Vaping (30d)	0.156	0.034 (0.028, 0.040)	0.122 (0.097, 0.145)	0.22 (0.18, 0.27)	10,095
Rx pain-med misuse (ever)	0.102	0.023 (0.018, 0.027)	0.079 (0.060, 0.099)	0.22 (0.17, 0.29)	10,440
Any illicit drug (ever)	0.119	0.024 (0.019, 0.029)	0.096 (0.075, 0.116)	0.20 (0.15, 0.25)	9,952
Binge drinking (30d)	0.103	0.015 (0.011, 0.020)	0.088 (0.069, 0.105)	0.15 (0.11, 0.20)	10,051

All ACME intervals exclude zero. The poor-mental-health mediator shows the same pattern with smaller shares (5–12% substances; 30% connectedness). Both H2 components are supported: distress mediates every pathway, and the mediated share for withdrawal (46%) is roughly double that of any substance outcome — withdrawal expresses distress; substance use additionally serves regulation functions two mood items do not capture.

5.8 Screening prototype

Table 8. Held-out test performance (bootstrap 95% CIs, 1,000 resamples). Selected XGBoost hyperparameters: max_depth 2, learning rate 0.010–0.017, 600–620 estimators (all targets favored shallow, slow-learning ensembles).

Target (prev.)	n	Model	AUC (95% CI)	PR-AUC (95% CI)	Brier
Any current substance (43.7%)	6,409	XGB-cal	0.694 (0.664–0.722)	0.636 (0.596–0.679)	0.218
		Logistic	0.697 (0.666–0.725)	0.641 (0.601–0.685)	0.217
Binge drinking (12.6%)	6,039	XGB-cal	0.693 (0.647–0.737)	0.287 (0.226–0.358)	0.103
		Logistic	0.691 (0.648–0.734)	0.268 (0.210–0.338)	0.103
Any illicit ever (13.9%)	6,139	XGB-cal	0.759 (0.718–0.797)	0.436 (0.361–0.513)	0.103
		Logistic	0.761 (0.721–0.799)	0.433 (0.358–0.509)	0.101

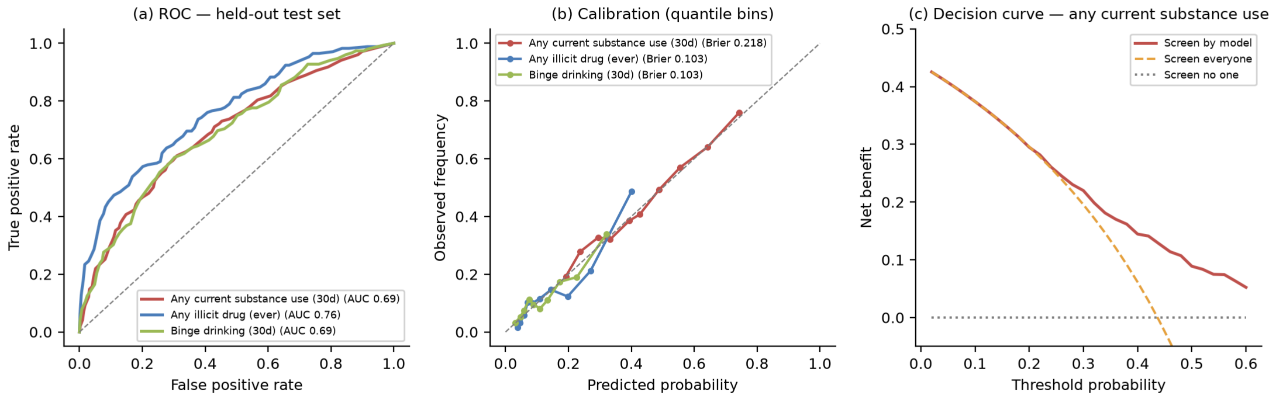

Figure 4. *Screening prototype, held-out test set: (a) ROC curves; (b) calibration in quantile bins against the identity line; (c) decision curve for any current substance use against screen-everyone and screen-no-one strategies.*

Calibration curves track the diagonal (Figure 4b), making the probabilities decision-grade. Operating points for any-current-substance: at threshold 0.24, sensitivity 90.5% with 48.2% precision (82% flagged); at the recommended threshold 0.32, sensitivity 81.3% with 51.7% precision against the 43.7% base rate (69% flagged; per 1,000 students: 455 true alerts, 426 supportive-outreach false alerts, 105 missed). Decision-curve analysis places model-guided outreach above both screen-everyone and screen-no-one throughout $p_t \in [0.05, 0.60]$.

Table 9. Fairness audit at the deployed operating point (threshold 0.32, any current substance use).

Group	n	Prevalence	AUC	Sensitivity	FPR	Precision
Female	657	0.47	0.688	0.842	0.688	0.524
Male	619	0.40	0.694	0.771	0.497	0.504
White	578	0.44	0.710	0.819	0.608	0.514
Multiracial (Hispanic)	225	0.44	0.695	0.800	0.640	0.500
Multiracial (non-Hispanic)	167	0.43	0.665	0.833	0.621	0.504

Subgroup AUCs span 0.665–0.710 (max gap 0.045) and precision is uniform (0.50–0.52). Sensitivity/FPR differences across groups (e.g., female students flagged more often, consistent with their higher exposure and prevalence) are reported openly; a single global threshold cannot equalize all group metrics simultaneously when base rates differ, and group-specific thresholds carry legal and ethical trade-offs addressed in Section 6.3. Because the logistic variant matches XGBoost within ~0.01 AUC everywhere, the deployment recommendation is the interpretable model, whose thirteen coefficients ship as published odds ratios (physical dating violence 2.12, witnessed community violence 1.68, parent substance problem 1.65, cyberbullying 1.44, sexual-violence exposure 1.41, basic-needs support 0.97).

5.9 Hypothesis summary

Hypothesis	Verdict	Key evidence
H1a Breadth	Supported	All 12 substance outcomes elevated; adjusted ORs 2.19–3.19, CIs exclude 1
H1b Severity gradient	Supported	PR rises 2.17 → 7.88 with substance severity
H2 Mediation	Supported	All ACME CIs exclude 0; withdrawal share (46%) ≈ 2× substance shares (15–23%)
H3 Dose–response	Supported	Strict monotonicity on every outcome, scores 0→3
H4 Predictability	Supported	CV AUC 0.70–0.79 substances; trauma features outrank demographics in SHAP

6. Discussion

6.1 Principal findings

All four pre-stated hypotheses were supported. The severity gradient (H1b) is the study's sharpest discriminating evidence: a dispositional risk-taking account predicts roughly uniform elevation across behaviors, while escalating trauma-driven coping predicts precisely the observed ordering — modest relative elevation for normative social substances, extreme relative elevation for substances used almost exclusively under severe regulation pressure. The mediation contrast (H2) sharpens the picture: withdrawal is predominantly a distress expression (46% mediated by one crude mood item), whereas substance use retains a large direct path, consistent with regulation functions (numbing, control, sleep) that mood items do not measure. Given single-item mediator measurement error, both mediated shares are plausibly underestimates.

6.2 Relation to prior literature

The adjusted ORs replicate the meta-analytic range (Hailes et al., 2019; Simpson & Miller, 2002) in a current, non-clinical, population-based adolescent sample. The dose–response results extend the ACE gradient (Felitti et al., 1998; Dube et al., 2003) to the 2023 cohort with items newly available at national scale. The predictive results align with prior YRBS machine-learning AUCs while using a deliberately impoverished, causally-disciplined feature set — evidence that most of the achievable signal is trauma burden itself, not behavioral correlates.

6.3 Implications and deployment ethics

Screening for victimization and household adversity — information many school health contexts already collect — could identify four of five students at elevated substance-use risk before behavior consolidates, at alert volumes decision-curve analysis shows are worth their cost whenever supportive outreach is the response. The mediation results specify the intervention: mental-health support, not discipline; a punitive response targets the symptom while deepening its cause. We are explicit that the prototype is not deployment-ready. Preconditions include prospective validation on the target population, IRB oversight, student and community consent, score confidentiality, routing exclusively to health professionals, and monitoring for structural-inequity encoding (Obermeyer et al., 2019). Excluding protected attributes from inputs while auditing on them is a minimum standard, not a resolution; when base rates differ across groups, no single threshold equalizes all error rates, and any group-specific correction must be weighed legally and ethically with the affected community.

6.4 Limitations

1. **Cross-sectional design.** Temporal order among exposure, mediator, and outcome is assumed, not observed; reverse pathways cannot be excluded; mediation relies on unverifiable sequential ignorability.
2. **Proxy exposure.** No age-at-victimization; non-disclosing survivors are misclassified as unexposed, biasing toward the null (associations are lower bounds).
3. **Single-item mediators** likely deflate mediated proportions.
4. **Self-report** with recall and social-desirability bias; out-of-school youth are outside the frame.
5. **Response rate.** The 35.4% overall response rate (49.8% school × 71.0% student) admits nonresponse bias that weighting only partially corrects.
6. **Approximate variance estimation** (PSU cluster-robust, ignoring strata) and subsample analyses inherited from questionnaire-version missingness; the screening models' complete-case restriction (n ≈ 6,000–6,400) may limit generalizability if intake completeness correlates with unmeasured risk.
7. **Unmeasured confounding** remains possible despite E-values of 2.3–3.0; candidate confounders (e.g., neighborhood disadvantage, genetic liability) plausibly operate below that strength net of included covariates, but cannot be ruled out.

6.5 Future directions

Pooling 2021 and 2023 cycles for replication and temporal-stability checks; formal fairness–accuracy frontier analysis with group-specific thresholds; ordinal/multistate models of polysubstance escalation; linkage-based longitudinal validation (Add Health-style) to convert mediation consistency checks into temporal tests; and prospective piloting of the screening instrument with school health partners under IRB oversight.

7. Conclusion

In the first national YRBS cycle to measure household ACEs, school connectedness, and parental monitoring alongside its violence and substance batteries, sexual-violence victimization and household adversity were associated with elevated use of every substance measured — with relative risk rising monotonically with substance severity — alongside reduced social connection and severe psychological distress that mediated a substantial share of both pathways. The pattern is consistent with trauma-driven coping rather than intrinsic risk-seeking. A calibrated, fairness-audited screening prototype using only intake-collectable information identified 81% of currently-using students at a decision-analytically supported operating point. Early, trauma-informed identification appears feasible with data schools already collect; the broadest-benefit intervention remains preventing the trauma.

Declarations

Data availability. Public-use CDC YRBS microdata (https://www.cdc.gov/yrbs/data/index.html). **Code availability.** Complete pipeline, analysis code, and dashboard source: github.com/eshalmin-basit/CSA-data-science-project. **Ethics.** Secondary analysis of de-identified public-use data; no IRB review required. **Funding.** None. **Competing interests.** None declared.

References

- Brener, N. D., Mpofu, J. J., Krause, K. H., Everett Jones, S., Thornton, J. E., Myles, Z., Harris, W. A., Chyen, D., Lim, C., Arbelo, M., & Mbaka, C. K. (2024). Overview and methods for the Youth Risk Behavior Surveillance System — United States, 2023. *MMWR Supplements, 73*(4), 1–12.
- Centers for Disease Control and Prevention. (2024). *2023 YRBS data user's guide*. U.S. Department of Health and Human Services.
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research, 16*, 321–357.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). ACM.
- Dube, S. R., Felitti, V. J., Dong, M., Chapman, D. P., Giles, W. H., & Anda, R. F. (2003). Childhood abuse, neglect, and household dysfunction and the risk of illicit drug use: The Adverse Childhood Experiences Study. *Pediatrics, 111*(3), 564–572.
- Felitti, V. J., Anda, R. F., Nordenberg, D., Williamson, D. F., Spitz, A. M., Edwards, V., Koss, M. P., & Marks, J. S. (1998). Relationship of childhood abuse and household dysfunction to many of the leading causes of death in adults: The Adverse Childhood Experiences (ACE) Study. *American Journal of Preventive Medicine, 14*(4), 245–258.
- Hailes, H. P., Yu, R., Danese, A., & Fazel, S. (2019). Long-term outcomes of childhood sexual abuse: An umbrella review. *The Lancet Psychiatry, 6*(10), 830–839.
- Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods, 15*(4), 309–334.
- Khantzian, E. J. (1997). The self-medication hypothesis of substance use disorders: A reconsideration and recent applications. *Harvard Review of Psychiatry, 4*(5), 231–244.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*, 4765–4774.
- Norman, R. E., Byambaa, M., De, R., Butchart, A., Scott, J., & Vos, T. (2012). The long-term health consequences of child physical abuse, emotional abuse, and neglect: A systematic review and meta-analysis. *PLoS Medicine, 9*(11), e1001349.
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science, 366*(6464), 447–453.
- Platt, J. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In A. J. Smola, P. Bartlett, B. Schölkopf, & D. Schuurmans (Eds.), *Advances in large margin classifiers* (pp. 61–74). MIT Press.
- Resnick, M. D., Bearman, P. S., Blum, R. W., Bauman, K. E., Harris, K. M., Jones, J., Tabor, J., Beuhring, T., Sieving, R. E., Shew, M., Ireland, M., Bearinger, L. H., & Udry, J. R. (1997). Protecting adolescents from harm: Findings from the National Longitudinal Study on Adolescent Health. *JAMA, 278*(10), 823–832.

- Simpson, T. L., & Miller, W. R. (2002). Concomitance between childhood sexual and physical abuse and substance use problems: A review. *Clinical Psychology Review, 22*(1), 27–77.
- Steiner, R. J., Sheremenko, G., Lesesne, C., Dittus, P. J., Sieving, R. E., & Ethier, K. A. (2019). Adolescent connectedness and adult health outcomes. *Pediatrics, 144*(1), e20183766.
- VanderWeele, T. J., & Ding, P. (2017). Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine, 167*(4), 268–274.
- Vickers, A. J., & Elkin, E. B. (2006). Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making, 26*(6), 565–574.